

CSE0612216: Database Management System Lab

Part A: Course Information

1. **Program:** BSc. in CSE
2. **Course Code:** CSE0612216
3. **Course Title:** Database Management System Lab
4. **Course Type (GED/Core Course/Electives):** Core Course
5. **Year/Level/Semester/Term:** 2nd year/ 1st Semester
6. **Academic Session:** Autumn 2026
7. **Section:** 3A2
8. **Prerequisite Course (If any):** N/A
9. **Credit Value:** 1.5
10. **Contact Hours:** 3.00 hours/Week
11. **Class Schedule:** Sunday (Room: 506, Period: 08.00 AM – 10.00 AM)
12. **Counseling Schedule:** Wednesday (9.00 AM – 12.00 PM),
Thursday (8.00 AM – 11.00 AM)
13. **Course Teacher:** Md. Moradul Siddique, Designation: Lecturer, Room: 418(10), Email: moradul_siddique@uits.edu.bd, Cell: +8801621291844, Specialization: Machine Learning, Deep Learning, Computer Vision, Cybersecurity, Bioinformatics, Internet of Things (IoT) and Healthcare AI.
14. **Rationale of the Course:** This course introduces the core principles and techniques required in the design and implementation of database systems. This introductory application-oriented course covers the relational database systems RDBMS - the predominant system for business scientific and engineering applications at present. It includes Entity-Relational model, Normalization, Relational model, Relational algebra, and data access queries as well as an introduction to SQL. It also covers essential DBMS concepts such as: Transaction Processing, Concurrency Control and Recovery. It also provides students with theoretical knowledge and practical skills in the use of databases and database management systems in information technology applications.
15. **Course Objectives:**
 1. To explain basic database concepts, applications, data models, schemas and instances.
 2. To demonstrate the use of constraints and relational algebra operations.
 3. Describe the basics of SQL and construct queries using SQL.
 4. To facilitate students in Database design.
 5. To familiarize issues of concurrency control and transaction management.
 6. To design a commercial relational database system (Oracle/MySQL) by writing SQL using the system.

Part B: Skill Mapping

16. Course Outcome (COs)

No.	COs	BT	CP/ WP	CA/ EA	KP/ WK	AT	DM&A
CO1	Illustrate a relational database model using an Entity–Relationship (ER) diagram and explain the fundamental elements of a Database Management System.	Cog 3	-	-	2,3	V, Q	Lecture, Demonstra tion and Discussion
CO2	By using modern tools, Apply the			-			Lecture,

	database design principles, SQL/PL SQL	Cog 4	-		3	LT, LF, CP	Demonstration and Discussion
CO3	Demonstrate the knowledge in projects with a relational database system and design a team-based project.	Cog 6	1, 3, 4	-	4,5,6	A, R, P	Lecture, Demonstration and Case Study

*COs- Course Outcome; BT- Bloom's Taxonomy; CP/WP- Complex Engineering Problems; EA/WA- Complex Engineering Activities; AT- Assessment Tools; KP/WK- Knowledge Profile; DM&A: Delivery Methods & Activities; LT- Lab Test, LF- Lab Final, Q- Quiz, R- Report, P-Presentation, V- Viva, A- Assignment, CP- Class Performance.

17. Mapping of COs with Program Outcomes (POs)

Course Outcomes (CO)	PO(a)	PO(b)	PO(c)	PO(d)	PO(e)	PO(f)	PO(g)	PO(h)	PO(i)	PO(j)	PO(k)	PO(l)
CO1	√											
CO2					√							
CO3			√									

Part C: Teaching Learning Approach

18. Course plan specifying contents, COs, co-curricular activities (if any), teaching learning and assessment strategy mapped with COs.

*Commencement of the semester (Autumn 2026): 22 July 2026 and Last class of the semester: 19 November 2026

Week	Topics	Suggested Activity and Teaching Strategy	Assessment Strategy	Corresponding COs
1	Basic of the Relational model concept, Key Constraints, DDL and DML Concepts.	Students design schemas, identify keys, and practice DDL/DML commands using examples, diagrams, demonstrations, and collaborative exercises for hands-on learning using XAMP.	Lab performance, Lab test	CO1
2	Basic SQL Operation in Database, Database CRUD Operations, the basic of DDL: Create DB, Drop DB, Create Table, Drop Table.	Students practice CRUD and DDL commands by creating and deleting databases and tables. Also uses demonstrations, hands-on labs, and small group tasks.	Lab performance, Lab test	CO1
3	The basics of DML: Select, insert, update, delete. Where, And, Or, Not, Order by, In, Between, Like etc.	Students practice DML queries using filters and conditions. Teaching uses real examples, guided SQL demos, and hands-on tasks to build query skills.	Lab performance, Lab test	CO1
4	Aggregate Function: count, min, max, avg, sum. Group By, Having.	Students apply aggregate functions with Group By and Having, then	Lab performance	CO1

	Manipulating the operations: Update, Delete	practice update/delete tasks. Teaching uses examples, live SQL demos, and guided exercises.	, Lab test	
5	Call for Project Proposal The basic structure of an SQL expression: Foreign Key and Referential Integrity, how to retrieve data from two tables, basic query operations on the table.	Students practice joins using foreign keys and apply referential integrity. Teaching uses examples, guided SQL demos, and small projects for multi-table query skills.	Lab performance , Lab test	CO3, CO1
6	Evaluation Test on basic SQL queries Changing the structure of the table: Alter	Students modify tables using ALTER and solve SQL quiz tasks. Teaching uses demonstrations, practice exercises, and short assessments to reinforce learning.	Lab performance , Lab test, Assignment	CO2
7	SQL Join: Natural Join, Inner Join, Left Join, Right Join, etc.	Students practice different SQL joins on sample tables. Teaching uses visual join diagrams, guided demos, and hands-on query exercises to build understanding.	Lab performance , Lab test	CO1, CO2
8	SQL Nested Queries, Subqueries and Views	Students write nested queries and create views. Teaching uses step-by-step demos, examples, and practice tasks to build advanced SQL query skills	Lab performance , Lab test	CO2
9	Normalization Administrative commands: Commit, Rollback, Savepoint	Students practice transactions using Commit, Rollback, and Savepoint. Teaching uses demonstrations, guided exercises, and error-based scenarios to reinforce transaction control concepts.	Lab performance , Lab test	CO2
10	Call for Project Proposal PL/SQL: Triggers, Menu design, NoSQL	Students create PL/SQL triggers and design simple menus. Teaching uses hands-on coding, step-by-step demonstrations, and mini-projects to enhance practical skills	Lab Discussion	CO3
11	Project ER diagram Functions: Number, Character, Conversion, Date Database with REST API	Students design ER diagrams and practice SQL functions on sample data. Teaching uses demonstrations, guided exercises, and mini-projects for practical understanding.	Lab performance , Lab test	CO3
12	Project Schema diagram Setting up a local development environment using PHP and MySQL	Students create schema diagrams and set up PHP-MySQL environments. Teaching uses guided installations, demonstrations, and hands-on exercises for practical web database development skills.	Lab performance , Lab test	CO3
13	Project Demonstration	Students present their database projects. Teaching uses demonstrations, peer review, and Q&A sessions to evaluate understanding, functionality, and practical implementation skills.	Presentation , Viva	CO1, CO2, CO3

14	Lab Final & Quiz	Students complete a final lab and quiz on SQL concepts. Teaching uses practical exercises, timed assessments, and interactive review for skill evaluation.	Lab test, Quiz	CO1, CO2, CO3
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Part D: Assessment Approach

19. Assessment and Evaluation

Assessment components		Marks Distribution
Continuous Assessment (CIA) (100%)	Class Attendance (10) Class Performance (10)	20%
	Lab Performance Lab Test 1 (10) Lab Final (20)	30%
	Quiz (20) Lab Report (10)	30%
	Project Presentation (10) Viva (10)	20%
	Total Marks	100%

20. Grading system

Numerical Grade	Letter Grade	Grade Point
80% and above	A + (A plus)	4.00
75% to less than 80%	A (A regular)	3.75
70% to less than 75%	A – (A minus)	3.50
65% to less than 70%	B + (B plus)	3.25
60% to less than 65%	B (B regular)	3.00
55% to less than 60%	B – (B minus)	2.75
50% to less than 55%	C + (C plus)	2.50
45% to less than 50%	C (C regular)	2.25
40% to less than 45%	D	2.00
Less than 40%	F	0.00

Part E: References

21. Learning Materials

- 1) **Recommended Readings:** Lecture notes and Necessary documents (Uploaded in Google classroom).

- 2) **Supplementary Readings**

Text Book:

1. Silberschatz, A., Korth, H. F., & Sudarshan, S. (2006). Database system concepts (5th ed.). McGraw-Hill.
2. Rob, P., & Coronel, C. (2003). Database systems: Design, implementation, & management (5th ed.). Thomson Course Technology.

3. Elmasri, R., & Navathe, S. B. (2005). Fundamentals of database systems (5th ed.). Addison-Wesley.

Others (Site):

1. SQL Tutorial, Retrieved from: <https://www.w3schools.com/sql/>
2. SQL Tutorial, Retrieved from: <https://www.tpointtech.com/sql-tutorial>
3. SQL Exercises, Practice, Solution, Retrieved from: <https://www.w3resource.com/sql-exercises/>
(Accessed 13 March, 2025)

3) **Others**.....(as applicable for the discipline/academic program): N/A

Part F: Additional Information

1. **Program Outcome:** <https://uits.ac.bd/public/cse>
2. **Outcome Based Curriculum:** https://uits.ac.bd/public/obe_curriculum_cse
3. **Course Catalog:** https://uits.ac.bd/public/course_catalog_cse
4. **Faculty Information:** <https://uits.ac.bd/public/faculty-members-cse>
5. **Assessments Rubrics:**
6. **Details of Bloom Taxonomy, Knowledge Profile, Complex Engineering Problems and Complex Engineering Activities:**
7. **Make-up Exam Procedure:**
8. **Attendance Policy:**
9. **Academic Calendar:** <https://uits.ac.bd/academic-calendar/>
10. **Academic Policies:**
11. **Proctorial Rules (Code of Conduct):** <https://uits.ac.bd/wp-content/uploads/2024/01/Code-of-Conduct.pdf>